Likelihood ratio tests

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Last time: Likelihood ratio test

Suppose we have a sample $\mathbf{X} = (X_1, ..., X_n)$ from some distribution with probability function $f(x|\boldsymbol{\theta})$, with parameter $\boldsymbol{\theta} \in \mathbb{R}^d$. We wish to test

$$H_0 : \boldsymbol{\theta} \in \Theta_0 \qquad H_A : \boldsymbol{\theta} \in \Theta_1$$

The **likelihood ratio test** (LRT) of these hypotheses rejects H_0 when:

$$\Lambda = \frac{\sup_{\boldsymbol{\theta} \in \Theta_0} L(\boldsymbol{\theta}|\mathbf{X})}{\sup_{\boldsymbol{\theta} \in \Theta} L(\boldsymbol{\theta}|\mathbf{X})} \leq k$$

choose to get desired type I error rate

$$\Lambda = \frac{L(\hat{\theta}_0 | X)}{L(\hat{\theta} | X)}$$

$\hat{\theta}_0 \in \Theta_0$ is restricted MLE of θ

$\hat{\theta} \in \Theta$ is unrestricted MLE of θ

Asymptotics of the LRT: scalar case

Suppose we have an iid sample $\mathbf{X} = (X_1, ..., X_n)$ from some distribution with probability function $f(x|\theta)$, with parameter $\theta \in \mathbb{R}$. We wish to test

$$H_0 : \theta = \theta_0 \qquad H_A : \theta \neq \theta_0$$

Theorem: Under H_0 (and assuming required regularity conditions),

$$-2\log(\Lambda) = -2\log\left(\frac{L(\theta_0|X)}{L(\hat{\theta}|X)}\right)$$

$$= 2\log L(\hat{\theta}|X) - 2\log L(\theta_0|X)$$

$$\xrightarrow{d} \chi^2_1$$

So:

1. choose desired significance level α
2. compute $-2\log\Lambda$
3. Reject H_0 if $-2\log\Lambda > \chi^2_{1,\alpha}$ ← upper α quantile of χ^2_1
4. p-value $= P(\chi^2_1 > -2\log\Lambda)$

Proof: WTS $2\log L(\hat{\theta}|X) - 2\log L(\theta_0|X) \xrightarrow{d} \chi^2_1$ under H_0

Let $\ell(\theta) = \log L(\theta|X) = \sum_{i=1}^{n} \log f(X_i|\theta)$

Taylor expansion of $\ell(\theta_0)$ around $\hat{\theta}$ (MLE):

$$\ell(\theta_0) \approx \ell(\hat{\theta}) + \underbrace{\ell'(\hat{\theta})}_{=0 \text{ (b/c } \hat{\theta} \text{ is MLE)}}(\theta_0 - \hat{\theta}) + \frac{1}{2}\ell''(\hat{\theta})(\theta_0 - \hat{\theta})^2$$

(This works b/c under H_0, $\hat{\theta} \xrightarrow{p} \theta_0$)

$$\Rightarrow \quad 2(\ell(\hat{\theta}) - \ell(\theta_0)) \approx -\ell''(\hat{\theta})(\theta_0 - \hat{\theta})^2 = -\frac{1}{n}\ell''(\hat{\theta})(\sqrt{n}(\hat{\theta} - \theta_0))^2$$

① $$-\frac{1}{n}\ell''(\hat{\theta}) = -\frac{1}{n}\sum_{i=1}^{n} \frac{\partial^2}{\partial\theta^2}\log f(X_i|\theta)\Big|_{\theta=\hat{\theta}}$$

$$\xrightarrow[\text{(WLLN)}]{p} -E\left[\frac{\partial^2}{\partial\theta_0^2}\log f(X_i|\theta_0)\right] \qquad (\hat{\theta} \xrightarrow{p} \theta_0)$$

$$= \mathcal{I}_1(\theta_0)$$

② $$\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} N(0, \mathcal{I}_1^{-1}(\theta_0))$$ (asymptotic normality of MLE)

$$= \mathcal{I}_1^{-\frac{1}{2}}(\theta_0) \cdot N(0,1)$$

$$(\sqrt{n}(\hat{\theta} - \theta_0))^2 \xrightarrow{d} (\mathcal{I}_1^{-1/2}(\theta_0) \cdot N(0,1))^2 \qquad \text{(CMT)}$$

$$= \mathcal{I}_1^{-1}(\theta_0) \cdot \chi^2_1$$

Slutsky's:

$$2(\ell(\hat{\theta}) - \ell(\theta_0)) \xrightarrow{d} \mathcal{I}_1(\theta_0) \cdot \mathcal{I}_1^{-1}(\theta_0) \cdot \chi^2_1 = \chi^2_1$$

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Asymptotics of the LRT: general case

Suppose we have an iid sample $\mathbf{X} = (X_1, ..., X_n)$ from some distribution with probability function $f(x|\boldsymbol{\theta})$, with parameter $\boldsymbol{\theta} \in \Theta \subseteq \mathbb{R}^d$. We wish to test

$$H_0 : \boldsymbol{\theta} \in \Theta_0 \qquad H_A : \boldsymbol{\theta} \in \Theta_1$$

Let $d = \dim(\Theta)$ be the number of free parameters specified by Θ, and let d_0 be the number of free parameters specified by Θ_0.

Theorem: Under H_0 (and assuming required regularity conditions),

$$-2\log \Lambda = -2\log\left(\frac{\sup_{\theta \in \Theta_0} L(\theta|X)}{\sup_{\theta \in \Theta} L(\theta|X)}\right) \xrightarrow{d} \chi^2_{d-d_0}$$

Ex: Regression w/ coefficient vector $\beta \in \mathbb{R}^p$

$H_0: \begin{pmatrix} \beta_2 \\ \beta_3 \end{pmatrix} = 0 \qquad H_A: \begin{pmatrix} \beta_2 \\ \beta_3 \end{pmatrix} \neq 0$

i.e. $\beta \in \Theta_0 = \{\beta : \beta_2, \beta_3 = 0\}$

i.e. $\beta = (\beta_0, \beta_1, 0, 0, \beta_4, \ldots, \beta_{p-1})^T$

$d = p$

$d_0 = p-2$

$d - d_0 = 2$

$\Rightarrow \chi^2_2$